

Space Exploration: The Obligation We can't Defer

Humanity carries questions that no single generation can resolve: our origins, our position in the universe, and our survival as a civilization. Space exploration is the latest and most expensive approach to investigate them, forcing the collision between the immediate need of people to survive and the civilizational debts owed across centuries. For a single lifetime, the border segregating necessity and indulgence seems intuitive; however, this distinction becomes controversial in endeavors requiring significant resources and risks.

Space exploration is a comprehensive term that encompasses the varied investigations into the universe beyond Earth's atmosphere by means of crewed and uncrewed spacecraft. Since Konstantin Tsiolkovsky's rocket studies in 1900, human beings have attempted to access the infinite darkness around Earth. These investigations have facilitated a more profound understanding of humanity, our genesis, our place in the universe, as well as our ultimate survival. Meanwhile, magnificent economic expense and enormous risk have been inevitable for every mission, even as the results are never certain. Therefore, this raises the question, is space exploration truly worth the costs and sacrifices it demands, or whether it is an unavoidable path for human development, or an indulgence for a species on a planet that needs our attention?

This essay argues that the significance of space exploration depends on the timeframe: what appears indulgence for a single generation becomes a civilizational obligation across time. The failure to recognize this urgency is the greatest challenge humanity is facing.

The Cost

Humanity's first steps into the universe promised not only remarkable achievements, but also enormous sacrifice. Following WWII, President Kennedy promised a man on the moon, "approving an immediate 89% increase in NASA" which ballooned "another 101%" the following year for the first lunar landing project, the Apollo Mission (Logsdon, "A Space Legacy" 29). NASA's mission to the moon was the "most expensive peacetime mobilization of U.S. financial and human resources ever under-taken in pursuit of a specific goal" (Logsdon, "A Space Legacy" 29). In 2025, NASA's budget still reached a budget of \$25.4 billion while China spent approximately \$12–14 billion, Europe nearly €7.8 billion, and India about \$1.6 billion; therefore, the national budgets on space reached around \$137 billion in total (Novaspace). While these extraterrestrial ambitions cost billions, it is morally distorted that millions of people are lacking food, health care, and habitats.

Even as we weigh the economic cost of space travel, physical sacrifice is involved in leaving Earth. On January 28, 1986, the space shuttle, Challenger, exploded shortly after its launch from Kennedy Space Center, claiming the lives of all seven astronauts. Less than 20 years later, on February 1, 2003, the Space Shuttle Columbia disintegrated as it reentered the Earth's atmosphere over Texas and Louisiana, causing the deaths of all seven crew members as well as the damage of data collections. Those two disasters not only meant the tragic loss of life, but also represented the forfeiture of human expertise, financial

investment, and scientific research causing some critics to lament that “unexpected benefits of the space program have been few and far between”(Bainbridge 516). These costs appear overwhelming without considering the timeframe against which they must be measured.

Life on Earth

Even as many argue that space exploration is not worth the risk or the enormous cost, the practical applications of exploring space are significant for the current global economy and international order. According to a 2011 study by Hugo van Manen, a Dutch strategic analyst, “6-7% of Western countries’ GDP (more than €800bn in the EU alone) and 11 out of 16 critical industries are dependent on the system for positioning, timing and navigation (PNT) to function”(van Manen 4). Furthermore, this reliance encourages space mining as a potential approach to decelerate Earth's environmental degradation. Although it sounds lavish and speculative, “the transition of mining from Earth to space could allow for continued growth of metal use on Earth while limiting environmental and social costs” (Corrado 6). In this sense, space exploration is an effective mechanism making the Earth environment tractable.

The Human Story

Beyond the cost and risk, space exploration is fundamental for our perennial attempt to understand the human story. As a species, outside of the “rise and fall of political system, the shift and change of ethical norms,” one instinct persists: “humanity is continually concerned with its (which is to say our) origins” (Milligan, 6). The empirical questioning of our origins through stories or beliefs need to be dated all the way back to the very beginnings of recorded human history. In Ancient China, it was a snowfall. In Babylon, it was a garden. In every culture, humans have grappled with their origins. Though modern science is less intuitive than ancient myths, it is indispensable in unraveling the start of humanity. The instinct is not indulgent; it is the mechanism for civilizations to orient themselves across time.

Genesis

“Tell me how the Universe came about, and I will tell you who you are” (Von Foerster, 2003, Milligan, 293),

Scientists like Charles Darwin practiced a specialized form of reflection that “humans have engaged in for at least tens of thousands of years” (Milligan, 6), attempting to uncover the origin of the species. As the Galapagos Islands are connected to our story, so are the heavens. Prehistoric people gazed at celestials, carved what they saw on rocks and built monuments like Stonehenge to mark lunar positions. Between 2000-300 BCE, Babylonians created astronomical diaries, discovered repeating planetary patterns, and developed mathematical calculations for planets. All these disciplined observations are not demanded by their survival, but triggered by the deeper questions of their position and identity in the universe.

The Scientific Revolution (1500-1700) gave figures like Nicolaus Copernicus, Galileo Galilei, Johannes Kepler, and Isaac Newton more advanced instruments, including the telescope, which deepened the question of who we are. Meanwhile, the debates between early

scientists—the physical identities of the universe and the authority of evidence over superstition — transformed astronomy into modern science, establishing the fundamentals for space exploration to eventually carry humanity beyond Earth's atmosphere entirely.

After World War I, scientists still believed the Milky Way contained the whole universe, until “Hubble's observation of far stars in the Andromeda Galaxy revealed the universe to be far larger than previously predicted” (Big Bang Britannica). Meanwhile, Georges Lemaître, a Belgian cosmologist suggested the big bang theory— “a single monumental explosion so great that all of the universe's matter and energy had once emerged from a single point long ago”(Big Bang Britannica). This theory assumed an enormous space explosion created hydrogen and energy which formed stars, and subsequent nuclear fission in stars created elements necessary for human beings. Although there is sufficient evidence to prove that the existence of an explosion triggered the emergence of all the mandatory elements for life, the cause of the big bang is still unknown.

Therefore, the question of our origin is not one that a single generation could answer. Every past civilization that gazed at the sky inherited the unsolved question. Space exploration is not merely a tool to corroborate the big bang theory but a moment in human history where civilizational curiosity finally meets the means to satisfy it. To dismiss space exploration as indulgence misinterprets it as the curiosity of a single generation, neglecting the obligation accumulated through human history that humanity is constantly pursuing.

Where is Everybody?

Space exploration is not exclusive for any single generation. The question of our place in the universe is as old as humanity itself. In 1950, “Enrico Fermi postulated his famous paradox: If life is common enough in the universe to give rise to us, then where is everybody” (Astronomy)? Fermi's quip, rooted in the cosmological principle that Earth is not in a privileged position in the universe, underlines another fundamentally human instinct: to question our place in an otherwise silent galaxy. Our unique existence has been the subject of centuries of speculation and exploration.

Philosopher Wolfhart Totschnig contradicted the foundational assumption of Fermi paradox, “the assumption that an intelligent species will spread itself as far and wide as possible” (Sutter). He argued that emergence of culture differentiates intelligent civilizations with simple organisms. Since habitable planets are scarce, intelligent civilizations would prefer concealment over expansion, “developing cultural tools to prohibit the expansion” (Sutter). The Drake Equation, the approach to estimate the number of civilizations around, has similar limitations. The equation has been debated since the 1950s when Frank Drake introduced the equation, a set of multiplicative factors as “only the first variable of the equation could be reasonably estimated from physical evidence or extrapolated from a statistically meaningful sample, the number of solar-type stars; others remain mysteries” (Smith).

The indeterminacy of the Drake Equation and the competing interpretations about Fermi's paradox reveal the limitation of armchair reasonings. Through the discovery of more exoplanets and data samples, scientists are able to yield more reliable approximations about whether, in fact, we are alone in the universe. Thus, space exploration is the means to

unravel the question, “whether we are alone”, and to define humanity’s obligations across civilizational time.

Our Death

Despite the cognitive value of space exploration, a “non-market benefit” of space exploration is the reduction of the risk of human extinctions” (D. Baum) from the perspective of the CBA. One threat is collision with Near Earth Objects (NEO). The developed technology resulting from space exploration projects was useful to defend Earth from NEO, for instance, asteroids or comets’ strikes. This is exemplified by NASA’s NEO Surveyor, human’s first space telescope currently under construction, to detect asteroids and comets that might become potential hazards to Earth.

Yet, the longer horizon is more consequential. Without dedication in space exploration, humanity would be difficult to sustain in the long term. The Sun is gradually but inevitably getting warmer. Within 500 million to one billion years, Earth’s temperature will surpass the maximum threshold for supporting life. Moreover, as the Sun is expanding in the Solar System, Earth will finally be pulled into the red giant and evaporate. Then, Earth could survive “only if the planet happened to be pulled out of the solar system by a star system that is passing by”(D. Baum), which demands levels of engineering sophistication that can only be achieved by space exploration. In the short term, it provides a hedge against immediate threats, including “nuclear warfare, pandemics, anthropogenic climate changes, and disruptive technology”(D. Baum). In these cases, self-sufficient space colonies that are established under the effort of space exploration will enable life to persist in the universe. All these efforts not only indicate the close correlation between space exploration and human survival, they also represent the shift of human roles from passive inhabitants on Earth to active agents who attempt to prolong life’s continuity in the universe.

Conclusion:

Ultimately, the question is not simply whether its cost outweighs the benefits, but whether humanity can be justifiably restrained from the pursuit of space exploration. Throughout human history, it is evident that every past generation has left the unanswered question to the next, each advancing humanity’s understanding but never completely. By rejecting the necessity of space exploration, it is not merely an act of restriction, but reckoning the limit of the Earth as the limit of human possibility. The distinction between indulgence and necessity will only exist if we confine our judgements to a single lifetime; the indulgence remaining within a generation will reveal itself as civilizational necessity across time, and humanity has never had more at stake in getting that distinction right.

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